



A Brief Historical Overview on the Science of Concurrent Aerobic and Strength Training

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Introduction

The history of our humankind is characterized by pioneers such as Marco Polo, Christopher Columbus, and Sir Edmund Hillary, all of whom have attempted to explore the limits of human existence. Also exercise scientists are driven by the desire to expand the boundaries of biological factors limiting physical performance. However, even though exercise has been part of humanity since the antiquity, it was not until the early twentieth century, that the first attempts were made to scientifically explore the biological processes underlying the adaptations induced by regular physical exercise. As a pioneer in exercise science, between 1910 and 1920 August Krogh developed the first devices to study in depth the physiology of physical activity. Among those were for example an electromagnetic bicycle ergometer and an apparatus for gas analysis, both of which allowed him to study the relative contribution of carbohydrates and fat as sources for energy of muscular contractions [1]. Krogh was also the first to show that capillaries are the drive for oxygenation during physical exercise [2], for which he was awarded the Nobel Prize in medicine and physiology a few years later. These first discoveries did also set off a number of other ground-breaking explorations, such as the Nobel Prize of A.V. Hill of England for his findings related to the production of heat in muscles in 1922 and the later work of Roger Bannister in the 1950s and Bengt Saltin in 1960 [3, 4], all of which are nowadays considered as the foundation of exercise science research.

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Also the science of progressive resistance training is a development of the twentieth century. Actually during World War II, most physicians still believed that strength training may be adverse to health and well-being, by sharing views of a well-known newspaper columnist stating that “extreme effort is not desirable in any kind of physical training nor is it good for the heart” [5]. However, only a few years later the first scientific papers of progressive overload resistance training were published by Thomas L. Delorme, who utilized progressive resistance training (defined as lifting multiple sets of the individual 10 repetition maximum) for the rehabilitation of injured servicemen [5]. Delorme’s book “Progressive Resistance Exercise: Technic and Medical Application” and his academic publications are nowadays understood as the foundation for the science of strength training and his research continued way beyond the official end of World War II. Apparently, the first publication utilizing strength training for fitness purposes appeared in 1946 in “The Physiotherapy Review” and showed that young women training 5 days a week would more than double their maximal strength within only 4 weeks—a finding which was truly ground-breaking at that time [6]. Thus, also during the subsequent years, research on strength training gained further popularity throughout the 1970s and 1980s, for example through pioneering research by Paavo Komi [7], as well as Michael H Stone, William J Kraemer, and Keijo Häkkinen in the 1980s [8, 9].

Concurrent Training and the “Interference Effect”

Interestingly, it was not until 1980 when the first studies of combining aerobic and strength training were published. Robert C Hickson discovered that the strength but not endurance development may be compromised when a high frequency of intensive running and strength training sessions (11 weekly sessions in previously untrained persons!) were performed concurrently for more than 6–8 weeks [10]. In fact, the conclusion of this study reads that it might be deleterious for strength athletes to perform strenuous endurance activities simultaneously. Even more remarking, Hickson also concluded that the compromised strength adaptations were not attributed to residual fatigue, despite the high volume of aerobic and strength training but the possible mechanisms behind these findings remained to be elucidated. Ever since his pioneering study, this phenomenon is known as the “interference effect.”

Already 5 years later, in 1985 Dudley and Djamil revisited the initial observations by Hickson by performing a study with a much lower training frequency (i.e., 3 weekly sessions of aerobic and resistance training, respectively) but also a much shorter training duration of only 7 weeks [11]. The main findings of this study were that the “interference effect” was only observed at high but not low angular velocities. While it was concluded that rapid force development might be more susceptible for concurrent aerobic training, it needs to be acknowledged that the duration of 7 weeks may have also been too short to show impaired maximal strength development because in the study of Hickson strength improvements started to plateau after 6–8 weeks in the concurrent training group [10].

The early conclusions on the incompatibility of aerobic and strength training were challenged about 10 years later, when the first papers on the effects of different concurrent training modes were published. It was hypothesized that residual fatigue induced by aerobic exercise may compromise the ability to develop tension during the subsequent strength loading [12, 13], when both exercises are performed in close proximity. In fact, it was speculated that the resulting compromised “quality” of the strength training (indicated by a lower absolute lifted load) may lead to impaired chronic neuromuscular adaptations, as observed in the initial study by Hickson [12]. Interestingly, in this study only minor lower body strength gains in previously untrained students were observed when running was consistently performed prior to strength training but no impairments were found in upper body strength, indicating the “interference effect” to be a local rather than a central phenomenon.

Around the same time, Sale and colleagues comprehensively investigated the effects of concurrent training performed on the same day compared to concurrent training carried out on alternating days by including muscle tissue sampling [14]. In their study, young men performed concurrent training twice a week for a duration of 20 weeks. While the magnitude of improvements in maximal strength was much larger in the group training on alternating days, no differences between the groups were observed for muscle hypertrophy and improvements in aerobic capacity. However, training on alternating days increased citrate synthase activity which might be beneficial for the total aerobic capacity and fractional utilization of maximal oxygen uptake ($\text{VO}_{2\text{max}}$).

In 1993, Collins and Snow [15] expanded on the findings of these earlier studies by comparing the physiological adaptations to two different exercise orders (i.e., aerobic followed by strength training and vice versa), which is nowadays understood as the “order effect” [16, 17]. In this study, both men and women performed either of the two exercise orders three times weekly, over a period of 7 weeks. It was found that both neuromuscular and cardiorespiratory adaptations appeared to have occurred independent of the exercise order following short-term training. Moreover, Collins and Snow actually did not observe differences in the training intensities (i.e., “quality”) of aerobic and strength exercises, irrespective of the training sequence [15].

A further development in the history of concurrent training was set only a few years later by a study of Häkkinen et al. [18], who confirmed the assumption that the training volume might be a crucial mediator in the magnitude of interference over a training period of 21 weeks [18]. While in this study no impaired maximal strength development was observed, the adaptations in rapid force development commenced to plateau already after 7 weeks of training, indeed confirming the observation from Dudley and Djamil in 1985 that explosive strength might possibly be more prone to the aerobic exercise-induced interference than maximal strength development [11]. In fact, this phenomenon remains a scientific interest until today [18, 19].

Yet another milestone was set concomitantly with the advances in the molecular science and the possibilities to look closer at the mechanisms underlying the potential interference in the signaling pathways during concurrent training. Among the

first proposed mechanisms behind the interference effect was the “AMPK-PKB switch” hypothesis presented by Atherton and colleagues [20]. This hypothesis was based on a model where rodents were electrically stimulated to mimic endurance and strength training. They proposed that AMPK activation by the endurance stimulus could ultimately inhibit mTOR activation and thereby reduce translation initiation and elongation [20]. This hypothesis was further developed in subsequent review papers, e.g., by Coffey and Hawley [21] and Hawley [22] and still remains one of the hypothesis being investigated, although evidence is emerging for this explanation representing an oversimplification [23].

Strength Training and Aerobic Performance

In his initial study, Hickson also concluded that there is little or no benefit for endurance athletes to strength train at the same time [10]. However, already in a paper published in the same year, the same group showed that untrained men nearly doubled time to exhaustion on the cycle ergometer after 10 weeks of pure strength training, without presenting significant changes in $\text{VO}_{2\text{max}}$ [24]. Because the effects were much smaller during treadmill running, it was concluded that improvements in cycling performance were predominantly attributed to increased maximal strength rather than changes in oxidative capacity. Moreover, the participants in this study were previously untrained and, thus, any type of training would have probably induced performance benefits.

In 1988, again Hickson expanded on his previous studies by incorporating strength exercise into the training routine of endurance athletes and concluded that at least endurance performance requiring fast-twitch fiber recruitment (i.e., short-term, anaerobic endurance performance) may actually be improved by strength training supplementation [25]. Moreover, no indications for impaired endurance performance were apparent and body mass and thigh girth were maintained (although this study lacked a control group performing endurance training only in the intervention period). Despite a lack of evidence for excessive increases in muscle mass, especially muscle hypertrophy is nowadays still of concern for endurance athletes and coaches, as increased bodyweight might hinder performance in weight-bearing disciplines, such as endurance running.

Throughout the subsequent years, the effects of strength training for aerobic performance were of minor research interest. However, in 1999 studies by Paavolainen et al. [26] and Hoff et al. [27] provided evidence for improved exercise economy and consequently overall endurance performance in cross-country skiers. Similarly to the early study by Hickson, also these studies showed that performance improvements occurred independent of improvements in $\text{VO}_{2\text{max}}$ and were much rather related to changes in neuromuscular characteristics. This is because in the study by Paavolainen et al. the improvements in 5 km running time were actually associated with a shorter ground contact times and, thus, a more economic running pattern.

Interestingly, even though only a few studies had investigated the effects of strength training for endurance athletes by the end of the last century, the majority

of these studies pointed towards favorable changes of endurance performance. However, in a textbook focusing on training and nutritional strategies for sport [28], this was reflected quite controversially. In the chapter “Training techniques for successful endurance performance,” the effects of strength training for sports such as swimming, rowing, cross-country-skiing, and endurance cycling were discussed. The conclusion for all of these endurance sports was that regular resistance training does not beneficially affect performance of endurance athletes but much rather may restrict the volume of “beneficial,” sport-specific training. Moreover, “The evidence against well-trained endurance athletes incorporating resistance training into their normal workouts to improve their endurance performance appears to be strong” ([28], pp. 136). While the early studies of Hickson and others were acknowledged, it was further suggested that “[...] for highly-trained athletes who are already capable of generating high power outputs in their chosen discipline, further improvements in strength are a less important factor in enhanced endurance performance. At the highest level of competition, increases in strength and power per se are not as critical to successful performance as the development of correct technique. The bottom line is that modern training studies do not support the use of resistance training programs for improving the performance of highly-trained athletes” ([28], pp. 137–138).

While this view is still shared by numerous coaches and athletes, about 20 years later a large number of studies have accumulated to provide evidence for strength training being an integral part of successful sports performance, also in endurance sports. Thus, it is about time to summarize the potential of regular strength training to improve athletic performance, thereby discussing the “does” and “don’t’s” of concurrent training prescription in order to convince sport practitioners, coaches, and athletes of the importance of combining aerobic and resistance training for optimal sports performance and health.

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